

Informal Regulation and Housing Supply: Evidence from Chicago’s Aldermanic Privilege

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Abstract

Land-use regulation in cities depends not only on the formal rules, but also on the individual politicians who administer them. Many U.S. cities allow substantial informal discretion in the development approval process, often exercised by the local politician representing the district where development is proposed. I study informal regulatory discretion in Chicago, where “aldermanic privilege” gives city council members substantial influence over development in their wards. I construct alderman-level “stringency scores” from permit processing times and use complementary event-study and boundary-discontinuity designs to estimate the effects of these informal regulations on housing supply and prices. Assignment to a more stringent alderman lowers permitting activity by 12%, while multifamily density is 14–18% lower immediately on the more-stringent side of a ward boundary. I also find that home prices are 4.6% higher on the more-stringent side, while rents are 1.7% higher but not statistically distinguishable from zero. These results show that informal regulation can distort housing supply and potentially affect citywide housing affordability, even in areas where the formal zoning rules are similar.

JEL Codes: R14, R31, R52

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1 Introduction

Housing costs in the United States have risen sharply, and a large literature points to land-use regulation as a central cause. Restrictive zoning, permitting frictions, and inclusionary zoning requirements limit housing supply, raise prices, and distort the allocation of labor across cities (Glaeser, Gyourko and Saks, 2005; Saiz, 2010; Turner, Haughwout and Van Der Klaauw, 2014; Hsieh and Moretti, 2019; Duranton and Puga, 2023; Soltas, 2024; Soltas and Gruber, 2026). However, formal zoning rules are not the only determinant of what gets built. Informal regulations, often exercised through the discretionary power of local politicians, also play a significant role in shaping development patterns. Many major cities in the United States have a fragmented regulatory approval system that includes the discretionary judgment of the local city councilmember or a similar politician.¹ This potential for individual discretion to differentiate otherwise similar regulatory environments can induce uncertainty in the development process and potentially lower housing supply in cities (Mayer and Somerville, 2000*a*; Hilber and Robert-Nicoud, 2013; Khan, 2021; Mast, 2024). Informal discretion from individual politicians can substantially lengthen the development timeline. When developers face long and unpredictable timelines, projects shrink, marginal investments become unviable, and housing supply falls (Cunningham, 2006; Bloom, 2009; Oh, Yang and Yoon, 2024). Despite this evidence, there is little empirical work on how individual politicians use these informal regulatory powers to shape housing supply.

This paper studies this form of informal regulatory power in the context of Chicago’s “aldermanic privilege,” the unwritten but influential authority of each of Chicago’s 50 city council members (aldermen) to unilaterally block or approve development in that member’s ward. Chicago has no city charter, so there is no statutory basis for this power (McClinton, 2025). It operates through an unspoken mutual understanding: other aldermen vote with the local member on land use matters in exchange for similar support in their own

¹Analogous forms of council-member deference in land-use decisions have been documented in Philadelphia, Los Angeles, and New York City (Burnett and Kogan, 2014; Jacobson, 2023; Mitre-Becerril and MacDonald, 2024; New York City Charter Revision Commission, 2025).

wards, city agencies follow the local alderman’s lead, and developers understand that major projects require the alderman’s blessing (Thale, 2005; Einstein, Glick and Palmer, 2019; Khan, 2021). This creates sharp changes in regulatory authority at ward boundaries, even when the neighborhood and zoning are quite similar. Two parcels on opposite sides of the same street, in the same neighborhood, with the same zoning, can face very different regulatory regimes depending on which alderman represents them. I use this sudden change in regulatory authority to study how informal regulations affect housing supply and prices.

To measure the regulatory environment each alderman creates, I construct “regulatory stringency scores” from processing times for permits likely to involve substantial aldermanic discretion, which include new construction, renovations, and demolitions. I deem these “high-discretion” permits since they are the most likely to require aldermanic approval. The motivation for using processing time variation is simple: delay makes projects less valuable, and a developer who expects years of meetings, redesigns, and uncertainty may build a smaller project or be deterred from building at all (Kydland and Prescott, 1982; Mayer and Somerville, 2000*a*; Paciorek, 2013; Gyourko, Hartley and Krimmel, 2021; Soltas and Gruber, 2026).

I first use the 2015 ward redistricting to test whether permitting changes when a census block is reassigned to a different ward. This design holds location fixed while changing the assigned alderman. Using this within-census-block variation, I find that reassignment toward greater stringency reduces the annual number of high-discretion permits recorded by the city by about 12% in the five years following the remap. This result is driven asymmetrically by movement toward more stringent aldermen: these census blocks see a 20.6% reduction in high-discretion permits, compared with a 6.9% increase for blocks reassigned to a more lenient alderman. A relatively precise null result for low-discretion permits shows that these effects operate through discretionary authority rather than general ward-level differences in bureaucratic efficiency.

I then look at downstream housing completions to estimate the effects of this informal

regulatory discretion on the density of realized housing supply. Following Bayer, Ferreira and McMillan (2007) and Kulka, Sood and Chiumenti (2026), I employ nonparametric regression-discontinuity designs at ward boundaries to estimate effects on new construction density. I find a 14% reduction in FAR and an 18% reduction in DUPAC on the more-stringent side of ward boundaries, with slightly larger effects for multifamily new construction.

I finally examine whether housing costs are higher in the places where the supply evidence predicts they should be. I find a sharp increase in home prices across the boundary of about 4.6%. I also find a 1.7% increase in listed rents, but the estimate is imprecise and not statistically significant. Although these results are more suggestive, their direction is consistent with lower supply on the more-stringent side and with informal regulation affecting affordability.

This paper contributes most directly to work on local political control and fragmented land-use authority. Fischel (2009) argues that homeowners use local institutions to restrict development and protect property values, and subsequent work shows how decentralized jurisdictions can adopt restrictions that are locally rational but collectively costly (Hilber and Robert-Nicoud, 2013; Ortalo-Magné and Prat, 2014; Monarrez and Schönholzer, 2023; Mast, 2024; Bordeu, 2025; Kulka, Sood and Chiumenti, 2026). I show that politicians within a fragmented system can create different regulatory environments even when formal rules are shared.

In Chicago specifically, Freemark (2020) studies formal upzoning and Khan (2021) finds that aldermen approve smaller rezonings near ward boundaries where the benefits to their constituents are lower. I add to this literature by showing that the effects of informal regulatory discretion extend beyond formal zoning or statutory upzonings and appear in permitting activity and the density of new construction, not only in zoned density.

The paper also contributes to work on permitting delays and time-to-build, since the effects of these informal regulations are often realized through development uncertainty and delays. Development lags are a first-order determinant of housing supply (Mayer and

Somerville, 2000*a,b*; Paciorek, 2013). Recent work finds that rules-based planning shortens construction timelines and that developers place substantial value on lots with ready-to-issue permits (Lyons and Sweeney, 2025; Soltas and Gruber, 2026). I use alderman-specific variation in permitting time and connect this to both permit activity and completed housing density.

2 Institutional Background and Data

2.1 Aldermanic Privilege and the Development Process

Chicago’s City Council consists of 50 aldermen, each representing a single ward. Chicago’s uniquely informal norms give each alderman broad authority over development within ward boundaries. This practice, known as “aldermanic privilege” or “aldermanic prerogative,” rests on custom rather than formal law, in part because Chicago has no city charter (McClintock, 2025). On matters affecting a single ward, other aldermen defer to the local representative, and city agencies follow the alderman’s lead (Thale, 2005; Zhang, 2011). The convention is strong enough that aldermen have often been described as “little mayors” governing their wards as personal fiefdoms (Einhorn, 1991). One alderman described the system candidly: “I often liken the City of Chicago to a feudal system, where the mayor is sort of a de facto king. And each alderman is the lord of their individual fiefdom.” (Keefe, 2013)

The formal mechanism. Two reforms narrowed this authority in 2019. A new construction code repealed a rule that had generally required the Department of Buildings to notify the local alderman of permit applications and wait ten days. Mayor Lightfoot’s Executive Order 2019-2 separately directed city departments not to defer to aldermen when municipal law did not require it. Neither reform eliminated aldermanic influence: other notification requirements remained, departments continued to solicit aldermanic input, and aldermen retained City Council authority over zoning and other legislative approvals (City of Chicago,

2019; Mayor’s Press Office, 2019; Schutz, 2019). Lightfoot captured the problem directly: “If you’ve got 50 separate fiefdoms that you have to deal with on issues like affordable housing, or just getting a sign on your building, it makes doing business with the city of Chicago damn near impossible.” (Spielman, 2019*b*) The Metropolitan Planning Council made the same point from the developer side: “before a development even gets to the point of being indefinitely delayed, many never make it past ‘Go’ because of required aldermanic letters of support.” (Metropolitan Planning Council, 2019)

How prerogative shapes development. Aldermanic discretion enters well before formal permit issuance and continues after a project is filed.

Pre-application. Experienced Chicago developers know that projects requiring rezoning or smaller modifications need the alderman’s support before formal papers are filed. Although not legally required, aldermanic support is all but essential in practice: without it, a developer risks investing heavily in design, legal, and engineering work before facing rejection at the political stage. One anonymous Chicago architect described the dynamic: “To get the alderman, you have to get the neighborhood group, and that’s where the trouble comes in. It’s fraught with risk, even to have this conversation.” (Giangreco, 2024)

Rezoning. For projects requiring a change to the zoning code, the alderman’s role is effectively dispositive. Although zoning amendments formally require a City Council vote, the council “routinely shows unquestioning deference to local aldermen even in the absence of any articulated local concern” (Chase, 2023). Equally consequential is the alderman’s ability to preemptively *downzone* parcels to foreclose future development. Joseph Schwieterman, a DePaul University professor who studies Chicago land use, noted that the practice “creates real uncertainty for real estate developers when downzoning becomes ad hoc or driven by the desire of an alderman to settle scores.” (Gallun, 2019) Documented episodes include Ald. Brendan Reilly’s proposals to downzone a River North site where a developer had proposed a 60-story tower and a separate site to block a 615-room hotel, and a Pilsen downzoning

orchestrated by Ald. Solis that stripped a parcel of its residential designation to kill a planned 465-unit apartment project.

Permit processing. For projects that proceed by right under existing zoning, the alderman retains substantial informal influence over the Department of Buildings permit queue. The corruption case against Ald. Ed Burke illustrated the mechanism in an extreme form: federal prosecutors documented how his staff withheld routine permits—including a driveway permit for a fast-food restaurant—until the property owner directed legal business to Burke’s law firm (Spielman, 2019a). The case made visible what developers had long understood: in Chicago, permit timing is another lever aldermen can use to delay and suppress projects they oppose. In interviews, a former alderman described substantial use of the permit process to deter developers from trying to overrule the alderman on rezoning and planning matters earlier in the process.

A recent example. Developer Fern Hill began community outreach in 2021, proposing a roughly 500-unit mixed-income tower at a transit-accessible site in the 2nd Ward, with approximately 100 affordable units and an estimated \$2.5 million in annual property tax revenue. After four years and more than ten community meetings, Ald. Brian Hopkins announced in January 2025 that he would not advance the project, citing opposition from residents of neighboring high-rises. He was explicit about the source of his authority: “Ultimately I am the decision maker on this. I’m not going to hide from that. This is a zoning decision and in Chicago we have something called aldermanic prerogative.” (Achong, 2025) Within two weeks the developer resubmitted a 349-unit proposal, cutting 151 units (30 percent) and 101 feet from the tower; Hopkins then approved it (TRD Staff, 2026). The episode captures both margins this paper studies: the approved project is smaller than what the developer proposed (intensive margin), and the four-year ordeal signals to other developers the cost of attempting large projects in wards with active aldermanic gatekeeping (extensive margin).

Federal scrutiny. In 2023, the U.S. Department of Housing and Urban Development (HUD) concluded that aldermanic prerogative operates as a “local veto” that allows wards to block, deter, or downsize affordable housing, especially in majority-white areas (Chase, 2023); outside groups have made similar civil-rights arguments (Shriver Center on Poverty Law, 2019). The question of whether to reform or abolish aldermanic privilege remains a central issue in Chicago politics.²

2.2 Building Permits and the Permit Process

The City’s public data contain permits recorded by the Department of Buildings since 2006. Each record reports when the application began and when the permit was issued, but cannot capture projects abandoned before they progress far enough to generate a permit record. I use the universe of “high-discretion” permits from 2006–2022 that have valid geographic coordinates and nonnegative processing times. Online Appendix A.3 describes the separate sample used to estimate stringency scores in the event study. I divide permits into two groups based on how much aldermanic discretion they are likely to face.

High-discretion permits are permits for projects where aldermen are most likely to matter: new construction, renovation or alteration, demolition, porch construction, and reinstated permits.³ These are the permit types most likely to draw community opposition and face delays due to community meetings or direct aldermanic intervention. *Low-discretion permits* are all remaining permit types outside this high-discretion group, excluding signs. I use them as a placebo comparison group because they are generally less exposed to discretionary aldermanic intervention.

High-discretion permits take longer to process and vary more across aldermen: their mean processing time is 57.3 days, compared with 26.2 days for low-discretion permits (Figure 1). Processing time is negatively associated with ward-month permit volume for high-

²See <https://news.wttw.com/elections/voters-guide/2023/races/city-council>, where each city council candidate was asked specifically about their stance on aldermanic prerogative.

³Porch construction permits have received extra scrutiny since the 2003 balcony collapse.

discretion permits but weakly positively associated for low-discretion permits, consistent with the former capturing regulatory friction rather than general administrative capacity. Online Appendix Table C.3 reports the corresponding statistics.

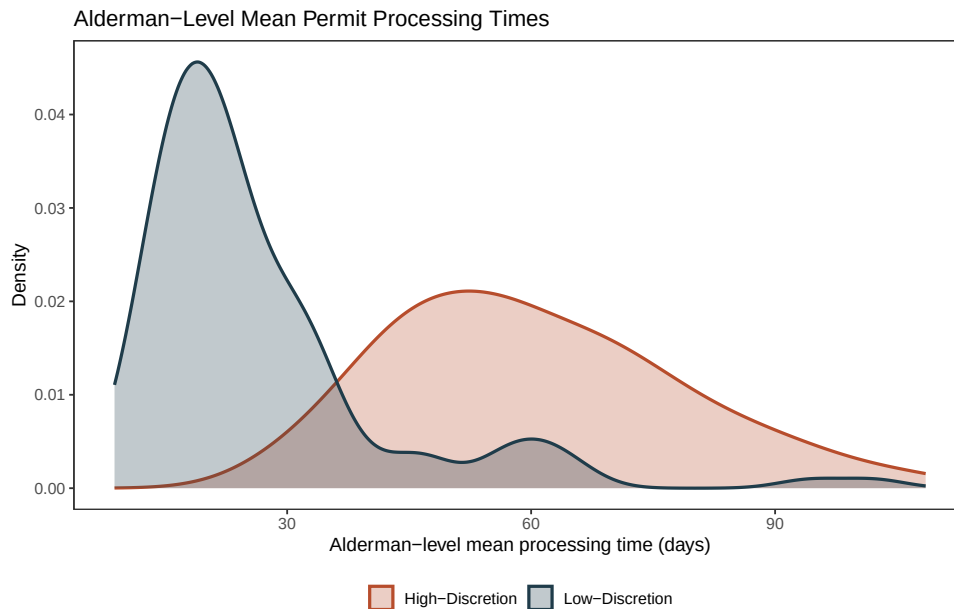


Figure 1: Alderman-Level Mean Processing Time for High- and Low-Discretion Permits

Notes: The figure plots the distribution of alderman-level mean permit processing times separately for high- and low-discretion permit groups, using permit records through December 2022. The high- and low-discretion permit groups are defined in Section 2.2. Processing time is days from application to issuance.

2.3 Regulatory Stringency Scores

I construct the stringency score using permit records from 2006–2022. I first adjust processing times for observed differences across wards, locations, permits, and months, then estimate the remaining alderman-specific differences. The first stage regresses log processing time at the permit level on ward-level demographic controls (median household income, racial composition, homeownership rate, population), permit-location geographic controls (distance to the central business district (CBD), lakefront proximity, Chicago Transit Authority (CTA) rail access), lagged ward-month permit volume, and fixed effects for year-month, permit type,

and review type:

$$\log(\text{Processing Time}_i) = X_i'\theta + \phi_{m(i)} + \psi_{c(i)} + \xi_{v(i)} + \varepsilon_i, \quad (1)$$

where i indexes permits, X_i contains the observed permit and location controls, and $m(i)$, $c(i)$, and $v(i)$ denote the permit’s year-month, permit class, and review type. Online Appendix Table B.1 reports the coefficients on the first-stage controls.

The second stage averages the adjusted processing times by ward and month, then regresses those averages on alderman indicators and the same lagged ward-month permit-volume control. Ward-months are weighted by the number of permits observed:

$$\bar{\varepsilon}_{wm} = \mu + \alpha_{a(w,m)} + \chi \text{Volume}_{w,m-1} + \nu_{wm}. \quad (2)$$

Here $a(w, m)$ denotes the alderman serving ward w in month m . The raw alderman fixed effects $\hat{\alpha}_a$ can be noisy for aldermen with few permits. I therefore apply empirical Bayes shrinkage (Kline, Rose and Walters, 2022, 2024) to pull imprecise estimates toward the grand mean, then standardize the final scores, denoted S_a , to have mean zero and unit standard deviation.

Online Appendix Figure B.1 maps the final stringency scores in 2014 and 2022, shows their distribution, and reports a low within-ward correlation between predecessor and successor aldermen ($\rho \approx 0.16$). The scores vary across wealthy and poor neighborhoods and across the North and South Sides, supporting an alderman-specific rather than ward-level interpretation.

2.4 New Construction Data

I construct a citywide dataset of new residential construction in Chicago between 2006 and 2022 using publicly available data from the Cook County Assessor’s Office. I combine residential improvement characteristics with commercial valuation records for properties assessed

under multifamily model groups. Together, these files report dwelling units, stories, building area, and land area. I locate each project using Assessor parcel records, assign it to a ward using City of Chicago shapefiles, and define the construction date as June 15 of the recorded year built since I cannot observe an exact date. For the boundary-discontinuity analyses, I divide ward boundaries into approximately 1,320-foot segments and assign each project to the nearest segment. Because some ward boundaries are several miles long, the segments ensure that compared projects are near both the same boundary and each other. I keep projects with positive lot area, positive building area, and at least one dwelling unit.

Because zoning measured today may reflect subsequent changes, I assign each project the broad zoning group in force on its assigned construction date. Online Appendix A.4 describes the historical zoning reconstruction.

For each project I compute two standard density measures: Floor Area Ratio (FAR), defined as total building area divided by lot area, and Dwelling Units Per Acre (DUPAC), defined as dwelling units divided by lot acreage. Distance to the nearest ward boundary is computed after projecting project locations and boundaries to the Illinois East State Plane coordinate system and is reported in feet.

Table 1 presents summary statistics. The estimation sample contains 3,692 new residential projects within 500 ft of a ward boundary, including 822 classified as multifamily. I report the multifamily sample separately because aldermanic discretion is likely to matter most for larger projects.

2.5 Rental Listings

RentHub provides scraped rental listings. I use Chicago listings from 2014 through December 2022 and collapse repeated scrapes to one floorplan-month observation, using the median asking rent observed that month. Rents are expressed in 2022 dollars using the monthly Chicago CPI-U.

Table 1: Summary Statistics for the Density Analysis Sample

	All New Construction	Multifamily
Average FAR	1.22	2.24
Average Units	6.71	26.28
Average DUPAC	29.47	69.91
Median Distance to Boundary (ft)	253	227
Ward/Segment Counts	124/849	87/415
N	3,692	822

Notes: Samples include new residential construction from 2006–2022 within 500ft of ward boundaries and use the same projects for the FAR and DUPAC estimates. Multifamily status follows the final project classification used in the density analysis.

2.6 Home Sales

Home sales come from Cook County Assessor records. I use valid market transactions of single-family homes and small multifamily buildings (2–6 units) from 2006 through 2022. Prices are expressed in 2022 dollars using the monthly Chicago CPI-U and winsorized at the 1st and 99th percentiles. Full 14-digit property index numbers (PINs) determine location and date-specific ward assignments, and same-year Assessor records provide property characteristics.

Online Appendix A provides additional detail on the rental and home sales data.

3 Conceptual Framework

3.1 Developer Problem

Delay makes projects less valuable. A developer who expects years of meetings, redesigns, and uncertainty may respond by building a smaller project, waiting, or walking away (Kydland and Prescott, 1982). The model below turns that intuition into a simple time-to-build problem.

Consider a developer choosing the scale $q \geq 0$ (units of housing) of a project on a given site. Construction costs are $C(q) = F\mathbf{1}\{q > 0\} + cq + \frac{\eta}{2}q^2$ with $F, \eta > 0$, so building requires

a fixed cost and marginal costs rise with project size. The developer can instead leave the site unbuilt and earn zero. Let P denote the local price level, taken as given at the project level.

Let S index the stringency of the local regulatory environment. Stringency maps into an *effective time-to-build* $T(S)$ with $T'(S) > 0$. The term $T(S)$ summarizes entitlement delay, redesign requirements, negotiation costs, financing frictions, and regulatory uncertainty that push expected revenue further into the future. If revenue arrives upon project completion, the developer's expected discounted revenue per unit is $e^{-rT(S)}P$, where r is the relevant discount or financing rate. The developer solves

$$\max_{q \geq 0} e^{-rT(S)}Pq - C(q).$$

Conditional on building, the first-order condition gives the interior scale

$$q^I(S, P) = \frac{e^{-rT(S)}P - c}{\eta}.$$

At this scale, the project's profit is $(e^{-rT(S)}P - c)^2 / (2\eta) - F$. The developer therefore builds only when the project covers its fixed cost, and the optimal choice is

$$q^*(S, P) = \begin{cases} \frac{e^{-rT(S)}P - c}{\eta}, & \text{if } e^{-rT(S)}P - c \geq \sqrt{2\eta F}, \\ 0, & \text{otherwise.} \end{cases}$$

For projects that are built, $\partial q^* / \partial T(S) < 0$. Longer or more burdensome regulatory processes therefore reduce project scale and can also push marginal projects from “build” to “do not build.”

3.2 From Developer Choices to Empirical Predictions

Indexing sites by i , total housing supply is $Q = \sum_i q_i^*(S_i, P_i)$. Higher $T(S_i)$ reduces Q through two direct margins: projects that are built at lower density and marginal projects that are no longer built. These are the main predictions I take to the data. The flow of permits requiring aldermanic review should fall when stringency increases (Section 4.1), and sites in more stringent regulatory environments should have lower completed density (Section 4.2).

With downward-sloping demand $P'(Q) < 0$, a supply contraction can also raise housing costs. That prediction is less sharp in the data because prices may adjust over a wider geography, but I find suggestive evidence that prices are higher on the more-stringent side of ward boundaries in Section 4.3.

3.3 Interpretation of the Stringency Score

In the model, these frictions enter through the index $rT(S)$. A higher value means revenue arrives later and is worth less today. That affects both the developer's intensive-margin density choice and the extensive-margin decision to build or not.

The aldermanic stringency score S_a constructed in Section 2.3 measures the alderman-specific component of observed processing time. I interpret a higher S_a as a more stringent regulatory environment and thus a higher effective time-to-build $T(S)$. Other dimensions of regulatory friction, such as impact fees, design mandates, or community benefit requirements, would enter as additional costs in $C(\cdot)$ and reinforce the same qualitative predictions. The core idea is that longer observed processing times for permits on larger discretionary projects are a noisy but informative signal of delay that lowers density, reduces aggregate supply, and may raise prices.

4 Results

4.1 Permit Activity After Redistricting

I first ask whether permit activity changes when the same location is assigned to a different alderman. Chicago’s 2015 ward redistricting moved some census blocks toward wards represented by more stringent aldermen and others toward wards represented by more lenient aldermen. Nearby blocks that remained in their original ward provide the comparison group. The design therefore holds the physical location fixed while changing its assigned regulatory environment. The primary analysis sample keeps boundaries for which both the origin- and destination-ward aldermen serving in 2014 remained in office when the new map took effect, since newly elected aldermen’s development preferences could be endogenous to prior development activity in the ward. I also require each block to have at least one high-discretion permit during the five years before redistricting. Without that restriction, any later permit on a block with no previous activity would mechanically appear as an increase.

The outcome is the number of high-discretion permits recorded on each census block in each year from 2010 through 2020. The outcome therefore measures when projects observed in the permit data first appeared in the process. These are the high-discretion permits defined in Section 2.2.

I estimate a Poisson pseudo-maximum likelihood event study that treats moves toward greater stringency and greater leniency as opposite changes:

$$\log \mathbb{E}[Y_{by} \mid \cdot] = \mu_b + \lambda_{g(b),y} + \sum_{t \neq -1} \beta_t Z_b \mathbf{1}\{y - 2015 = t\}. \quad (3)$$

Here Y_{by} is the permit count for block b in calendar year y , μ_b is a block fixed effect, and $\lambda_{g(b),y}$ is a ward-pair-by-year fixed effect for the ward pair $g(b)$ containing block b . Event time is $t = y - 2015$, with $t = -1$ omitted. Let ΔS_b be the destination-minus-origin difference in standardized aldermanic stringency for block b . For blocks that change wards,

$Z_b = \text{sign}(\Delta S_b)$; it equals zero for blocks that remain in the same ward. I classify moves using the 2014 incumbents in each block’s origin and destination wards, with scores estimated only from permits filed between 2006 and 2014. I do not redefine treatment using the 2015 election winners because those elections may themselves reflect local preferences toward development. This specification imposes equal-and-opposite effects in logs for assignments toward more stringent and more lenient aldermen. Panel A of Figure 2 plots the coefficients β_t . The pre-trends are not distinguishable from zero. The pooled post-period estimate is -0.127 ($\text{SE} = 0.049$), equivalent to an 11.9% reduction in the annual number of high-discretion permits associated with a move toward greater stringency. Panels B and C relax the symmetry restriction by estimating the two directions separately. The decline in permitting activity comes mostly from moves toward more stringent aldermen, which reduce high-discretion permits by 20.6%; blocks assigned toward more lenient aldermen experience a 6.9% increase. Online Appendix Table C.2 reports the corresponding pooled estimates without the symmetry restriction and shows the results are nearly identical.

This asymmetry suggests that aldermanic discretion operates through uniquely stringent aldermen blocking activity that would otherwise occur, rather than uniquely lenient aldermen allowing activity that formal city regulations would otherwise block. This distinction matters for the likely aggregate effect of informal regulatory discretion on housing supply because it is possible *ex ante* that a few particularly pro-development aldermen approve more housing than a centralized system would. The results instead suggest that informal aldermanic discretion reduces housing supply relative to a more centralized counterfactual.

4.2 Density at Ward Boundaries

I next examine the individual new construction projects described in Section 2.4 on opposite sides of the same ward boundary, moving the unit of analysis from census blocks to projects. Projects within 500 ft of the same boundary segment face similar neighborhood amenities and demand conditions, but different aldermen. Following Bayer, Ferreira and McMillan

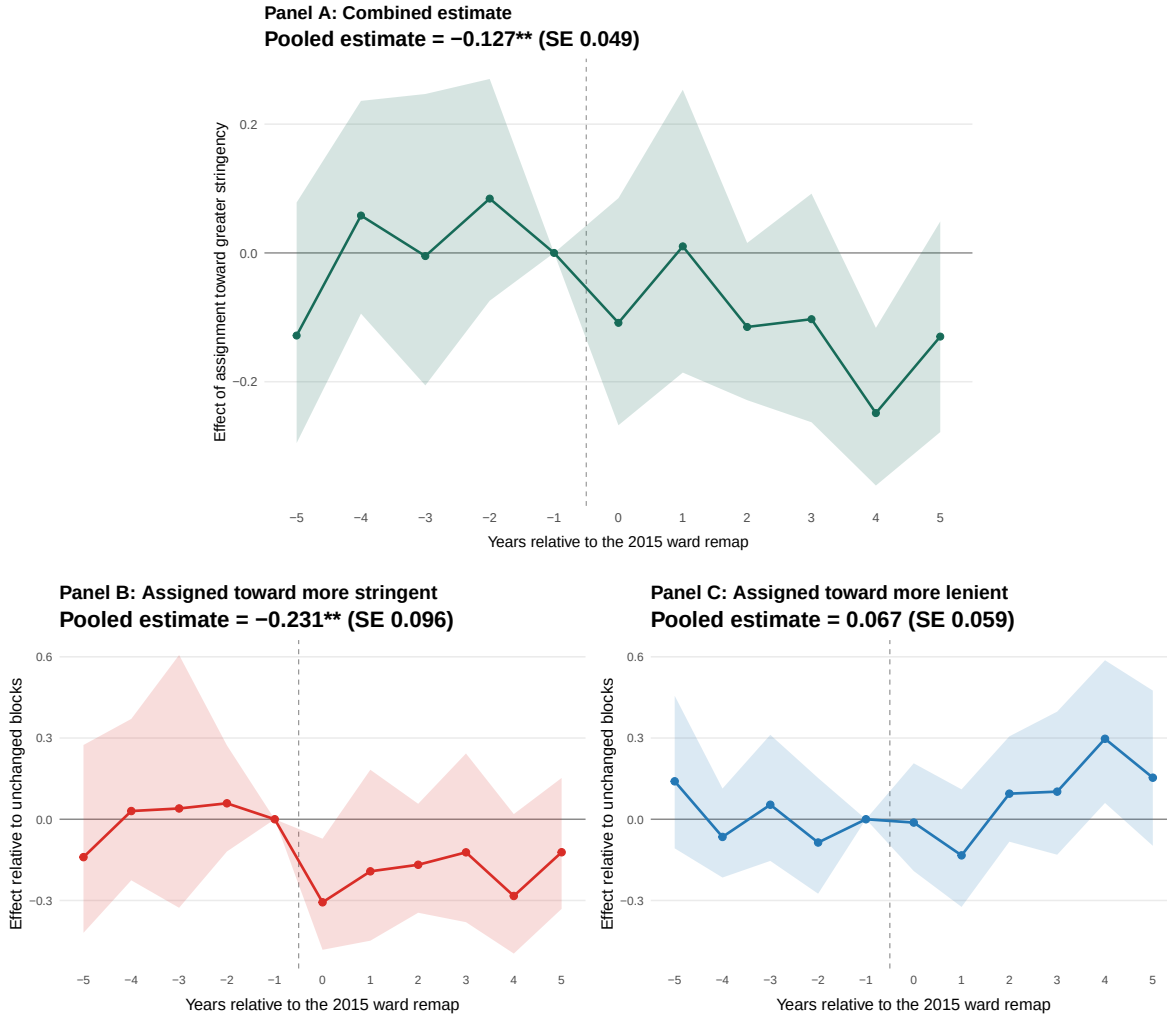


Figure 2: Permit Activity After the 2015 Ward Redistricting

Notes: Block-year Poisson pseudo-maximum likelihood event study for calendar years 2010–2020. The outcome is the number of high-discretion permits per block, grouped by application year. The sample includes blocks within 500 ft of ward boundaries. It retains blocks whose 2014 origin- and destination-ward incumbents remained in office when the new map took effect. Blocks must have at least one high-discretion permit during event times $t = -5$ through $t = -1$. Blocks are classified according to whether the 2015 map assigned them toward an alderman who was more or less stringent than the origin-ward alderman; unchanged blocks are the comparison group. Stringency scores are estimated from permits filed between 2006 and 2014. The model includes block and ward-pair $\times$ year fixed effects. Event time $t = -1$ is omitted. Panel A uses a reassignment variable equal to one for blocks assigned toward greater stringency, minus one for blocks assigned toward greater leniency, and zero for unchanged blocks. Panels B and C estimate the two reassignment directions separately relative to unchanged blocks. Each point shows the percentage change in permits relative to the year before redistricting and the subtitles report the pooled estimates in logs. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. The pooled coefficients are -0.127 (SE = 0.049) in Panel A, -0.231 (SE = 0.096) in Panel B, and 0.067 (SE = 0.059) in Panel C. The Panel A pre-trend test has $p = 0.394$.

(2007) and Kulka, Sood and Chiumenti (2026), I divide signed distance to the boundary into 100 ft bands and compare projects in each band with projects in the nearest band on the less-stringent side. This flexible specification does not require density to follow a linear or polynomial relationship with distance. I estimate

$$\begin{aligned} \log(Y_{isy}) = & \sum_{k \in \mathcal{K}, k \neq [-100, 0)} \kappa_k \mathbf{1}\{d_{is} \in k\} + X_i' \theta \\ & + \zeta_{z(i,y)} + \delta_s + \gamma_y + \varepsilon_{isy}. \end{aligned} \quad (4)$$

Here Y_{isy} is FAR or DUPAC for project i , assigned to boundary segment s and built in calendar year y . The running variable d_{is} is negative on the less-stringent side and positive on the more-stringent side. The set $\mathcal{K}$ contains ten 100 ft bands from -500 to 500 ft. The band from -100 to 0 ft is omitted, so $\kappa_{[0,100)}$ compares projects immediately across the boundary. The vector X_i contains the White population share, the Black population share, median household income, the bachelor's-degree share, and the homeownership rate on the project's side of the boundary. The model also includes the zoning group in force when the project was built, boundary-segment fixed effects, and construction-year fixed effects. Standard errors are clustered by ward pair.

Figure 3 plots the distance-band estimates. The omitted band is normalized to zero, and the subtitle in each panel reports $\kappa_{[0,100)}$. For all new construction, FAR is about 12% lower and dwelling units per acre about 16% lower immediately across the boundary on the more-stringent side. Among multifamily projects, FAR is about 14% lower and dwelling units per acre about 18% lower. All four differences are statistically distinguishable from zero at the 10% level or better. The similar results for FAR and dwelling units per acre indicate that the difference is present in both floor space and the number of homes placed on a parcel.

As a falsification test, I repeat the same comparison at artificial cutoffs 1000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority. Online Appendix Figure D.1 shows no corresponding break: none of the eight

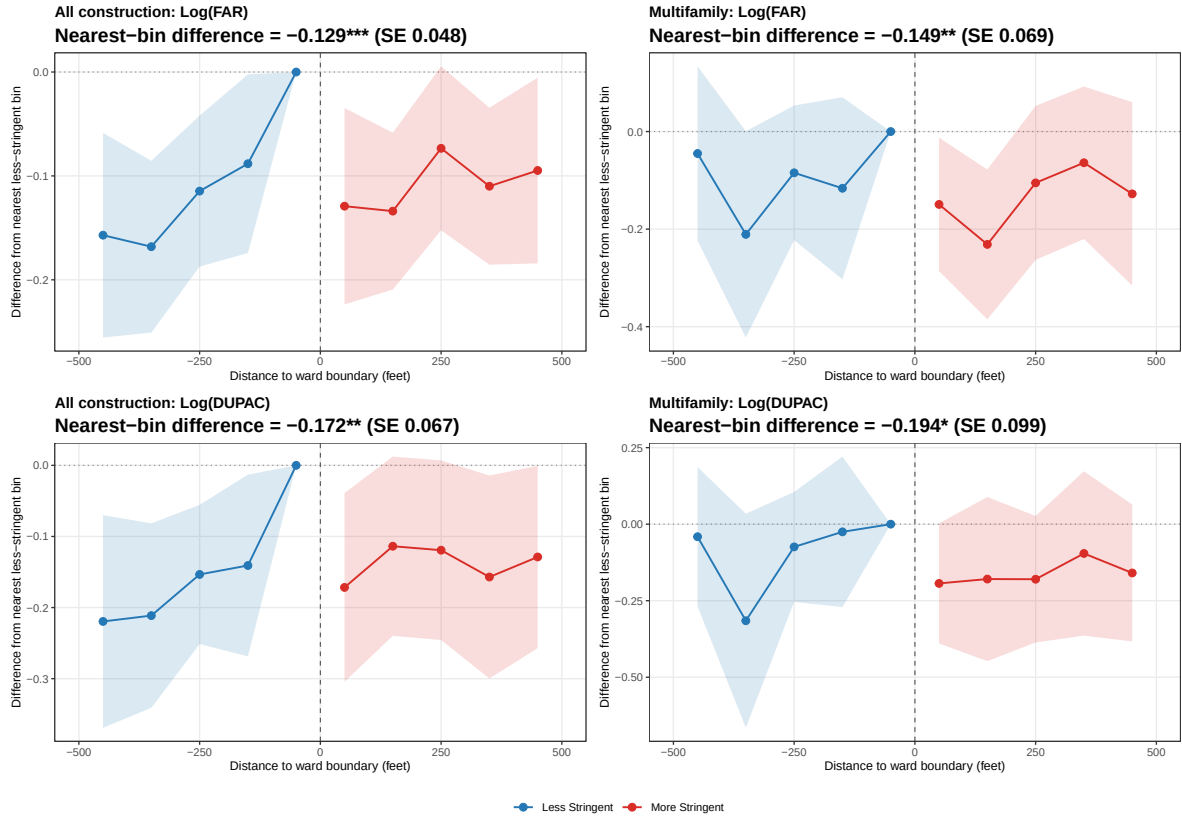


Figure 3: Development Density at Ward Boundaries

Notes: Left column: all construction; right column: multifamily. Top row: log(FAR); bottom row: log(DUPAC). Samples include new construction within 500 ft of ward boundaries. Each point is the coefficient for a 100 ft distance band from Equation (4). The nearest band on the less-stringent side, from -100 to 0 ft, is omitted and normalized to zero. Subtitles report the coefficient for the nearest band on the more-stringent side, from 0 to 100 ft. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. Negative distance indicates the less-stringent side and positive distance the more-stringent side.

nearest-band placebo estimates is statistically distinguishable from zero. The contrast with the true boundary makes it less likely that the main estimates reflect a general spatial pattern within wards. Online Appendix D presents the placebo figures along with continuity tests for fixed location characteristics, restrictions based on boundary geometry, and donut checks.

4.3 Evidence on Rents and Sale Prices

The permit and density results indicate that more stringent aldermen reduce development activity and the density of completed housing. The model in Section 3 indicates that this should feed into housing costs, the most directly policy-relevant outcome given Chicago’s current rent and price growth. For rents and sales, I estimate

$$\log(Y_{isp}) = \sum_{k \in \mathcal{K}, k \neq [-100, 0)} \kappa_k \mathbf{1}\{d_{is} \in k\} + X_i' \theta + \delta_{sp} + \varepsilon_{isp}, \quad (5)$$

where Y_{isp} is the real listed rent or real sale price for observation i near boundary segment s in calendar period p . As in the density analysis, $\mathcal{K}$ contains ten 100 ft distance bands and the band from -100 to 0 ft is omitted. The coefficient $\kappa_{[0, 100)}$ therefore compares observations in the two bands immediately across the boundary. The fixed effects compare observations near the same boundary segment in the same month for rents and the same quarter for sales. The controls include unit or property characteristics and distances to nearby schools, parks, major roads, CTA stops, and Lake Michigan. Standard errors are clustered by ward pair.

Figure 4 plots these distance-band estimates.

Listed rents are 1.7% higher on the more-stringent side relative to the less-stringent side, although the estimate is imprecise. Home sale prices are 4.6% higher on the more-stringent side relative to the less-stringent side. I treat the sales result as suggestive because transactions are less frequent near any particular boundary and their composition is harder to hold fixed. Online Appendix E shows that the result disappears at artificial cutoffs inside either ward and is similar when I restrict the sample to straight boundaries or exclude

observations immediately next to the boundary.

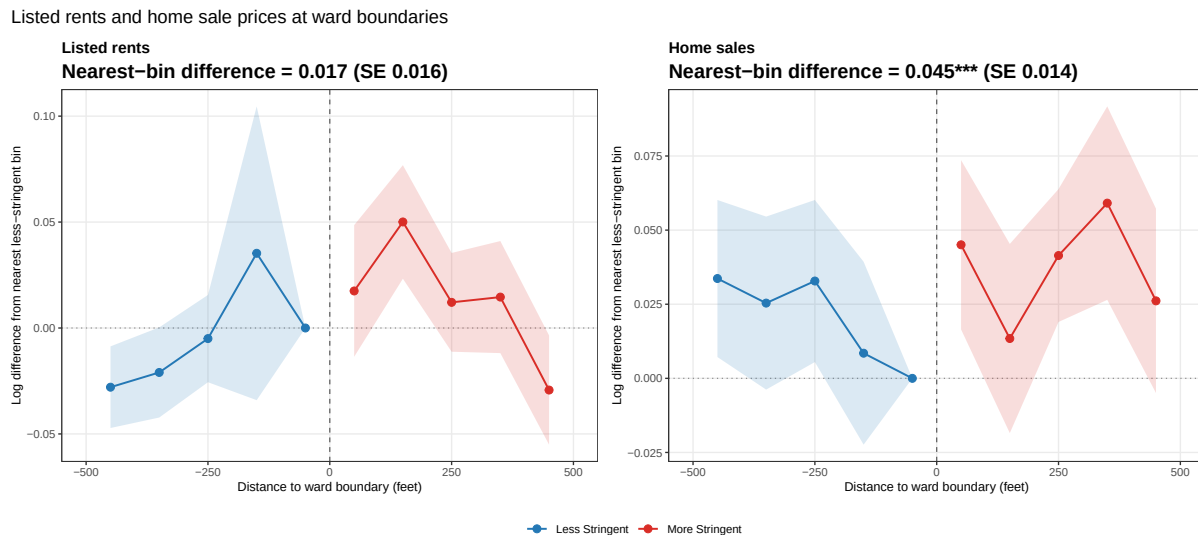


Figure 4: Listed Rents and Home Sale Prices at Ward Boundaries

Notes: Panel A uses listed-rent floorplan-month observations from 2014–2022 and segment-by-month fixed effects. Panel B uses filtered residential transactions from 2006–2022 and segment-by-quarter fixed effects. Both panels include housing characteristics, distances to nearby amenities, and the fixed effects described in the text. Each point is the estimated log difference between a 100 ft distance band and the band immediately inside the less-stringent side, which is normalized to zero. Shaded areas are 95% confidence intervals. Subtitles report the comparison between the two bands adjacent to the boundary. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Discussion

5.1 Interpretation and Mechanisms

Taken together, the redistricting and boundary results show that individual politicians create meaningfully different regulatory environments within a single city. Within-census-block variation shows that permitting falls when a census block is reassigned to a more stringent alderman, while completed projects on the more-stringent side of ward boundaries are built at lower density. The price and rent comparisons are more tentative, but their direction is consistent with these differences in housing supply.

The central implication of these results is that the effect of political fragmentation on housing supply extends beyond formal zoning rules. Two otherwise similar projects gov-

erned by the same municipal code can face different timelines, negotiations, and prospects for approval because they are represented by different aldermen. Political fragmentation can therefore arise within a city, even without separate municipal governments or formally different regulatory codes.

These informal regulations can affect development along both extensive and intensive margins. Additional meetings, redesigns, and uncertain approval timelines reduce the expected return to development. Developers may respond to these additional demands, or the threat of them, by shrinking or abandoning projects before they enter the permitting data. The Fern Hill episode described in Section 2 illustrates these margins. The project that ultimately received approval was smaller than the original proposal, and approval followed four years of meetings and negotiation. The time-to-build framework in Section 3 captures the underlying economic force: delay lowers the value of development, reduces the scale of projects that remain viable, and can cause marginal projects to disappear.

These findings complement Khan (2021), who shows that Chicago aldermen approve smaller increases in allowed density near ward boundaries, where their constituents capture fewer of the benefits from development. Informal political control can therefore affect not only what the zoning code allows, but also whether development proceeds and what ultimately gets built.

5.2 Informal Discretion Beyond Chicago

Chicago provides an especially clear setting because aldermanic privilege is unusually powerful, but the underlying institution is not unique to Chicago. Local council-member deference in land-use decisions has also been documented in Philadelphia, Los Angeles, and New York City (The Pew Charitable Trusts, 2015; Ryu, 2020; Anbinder, 2025; New York City Charter Revision Commission, 2025; Mitre-Becerril and MacDonald, 2024; Burnett and Kogan, 2014; Jacobson, 2023).

In many major U.S. cities, individual politicians can create distinct regulatory environ-

ments when projects require discretionary approval and other officials defer to the representative of the affected district. Under these conditions, the identity and preferences of the local politician can matter even when the written rules remain similar across districts. The broader lesson—that informal regulations can dampen housing supply and affect affordability—likely applies to other major cities as well.

5.3 Policy Implications

Local politicians may possess useful information about neighborhood conditions and provide residents with a clear channel for political accountability. But ward-level representatives also have little reason to internalize the citywide benefits of additional housing, and informal discretion allows them to impose delay and uncertainty after the formal rules have been established. The resulting regulatory environment may reflect the preferences of an individual officeholder or a narrow slice of constituents rather than a broad city welfare metric.

While this paper cannot pin down the optimal balance between local input and centralized authority, it does point to predictability and transparency as important goals for reform. Expanding the scope of by-right approval would allow projects that comply with established rules to proceed without obtaining the support of an individual alderman. Published approval criteria, enforceable timelines, and meaningful appeal procedures could preserve local participation while limiting uncertainty. Moving some decisions to citywide bodies could also make it easier to account for housing benefits that extend beyond the ward in which a project is located.

Chicago’s lack of a city charter makes the allocation of authority particularly dependent on unwritten norms. A charter would not by itself eliminate aldermanic influence, but it could formally define approval authority, procedures, and avenues for appeal (McClintock, 2025). More generally, reforms aimed only at zoning classifications or formal voting rules may have limited effects if individual politicians retain informal control over permitting, project design, or administrative implementation.

These institutional choices matter as Chicago confronts rapidly rising rents (Davis, 2026). More broadly, the results show why evaluations of land-use regulation should look beyond the written zoning code. When individual political actors possess substantial discretionary authority, who administers the rules can matter as much as the rules themselves.

6 Conclusion

This paper studies how informal regulations and the discretion of individual political actors affect housing supply within a city. I focus on Chicago’s aldermanic privilege, the informal power of each of the city’s 50 aldermen to block or approve development in that member’s ward. This institution allows individual politicians to create different regulatory environments within a single municipal system.

Using permit processing times for projects where aldermen are most likely to matter, I construct aldermanic stringency scores that isolate the alderman-specific component of regulatory delay after accounting for permit, location, time, and ward characteristics. Using the 2015 ward redistricting to identify changes in assigned stringency, I find that reassignment toward greater stringency reduces the annual number of high-discretion permits recorded by the city by about 12% in the five years following the remap. A boundary comparison shows that new construction is less dense on the more-stringent side of ward boundaries, with multifamily FAR about 14% lower and DUPAC about 18% lower immediately across the boundary. I also find a 4.6% increase in home prices and a suggestive 1.7% increase in listed rents, although the rent estimate is imprecise.

These results show that formal zoning rules do not fully describe the regulatory environment facing a developer. Similar projects governed by the same municipal and zoning codes can face different timelines and prospects for approval depending on the local politician. Effective land-use reforms must address not only the formal rules governing what can be built, but also the informal political institutions that determine whether and how development

proceeds.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used OpenAI's Codex to assist with manuscript editing, primarily for clarity and concision, and with coding for data cleaning and analysis. The author reviewed and validated all code and edited all manuscript content as needed and takes full responsibility for the content of the published article.

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A Data and Sample Construction

This appendix provides additional detail on the data sources and sample rules described in Section 2.

A.1 Data Sources

The property data come from Cook County Assessor sales, assessment, parcel, and improvement files. RentHub provides rental listings from 2014–2022. Building permits and ward boundaries come from the City of Chicago. The amenity controls use school locations, Chicago Park District boundaries, major streets, CTA stops, and Lake Michigan geometry; CTA distances use the permanent station network in place during the listing month or on the sale date.

Demographic controls come from American Community Survey block groups and 2010 and 2020 census blocks. I allocate population, households, tenure, race, ethnicity, and education counts to wards according to the share of each block group’s area falling within the relevant ward boundaries. Ward shares use those allocated counts, while ward median household income is the household-weighted mean of block group medians. When historical ward polygons overlap slightly, I scale the intersections to the block group’s area inside the union of Chicago wards so no area is counted twice.

A.2 High-Discretion Permits

The permit records cover Department of Buildings permits since 2006. The high-discretion group contains the permit types defined in Section 2.2; low-discretion placebo permits are the remaining routine types, excluding signs. The event-study outcomes omit records marked cancelled, revoked, or suspended.

Processing time is the number of days between application and issuance. The processing-time regression excludes missing dates, negative durations, and same-day permits; same-day

permits remain in the lagged ward-month permit count. I assign each permit to the ward containing its coordinates under the boundaries in force on its application date.

A.3 Permit Event-Study Panel

The treatment panel compares each census block’s ward under the pre-2015 and post-2015 maps. Blocks are projected to EPSG:3435 and assigned by largest-area overlap. Let $a_o(b)$ and $a_d(b)$ denote the aldermen representing census block b under the origin and destination ward maps. Blocks moved to a different ward are classified using the difference between these aldermen’s stringency scores:

$$\Delta S_b = S_{a_d(b)} - S_{a_o(b)}.$$

Positive values indicate assignment toward greater stringency, negative values indicate assignment toward greater leniency, and unchanged blocks form the comparison group.

For this design, stringency scores use permits filed from 2006 through 2014. Both origin and destination scores are assigned using the incumbents serving in 2014. The main sample retains blocks only when both incumbents remained in office when the new map took effect, so treatment matches the change the block experienced without using the outcome of the 2015 election. Online Appendix C.4 adds wards with electoral turnover and interprets the resulting estimate as assigned rather than necessarily realized stringency.

Permits are aggregated by block and application year for event times $t = -5$ through $t = 5$. The sample includes blocks whose polygon centroid is within 500 ft of a pre-redistricting boundary. Treated blocks use the fixed origin-destination boundary created by redistricting; comparison blocks use their nearest incident pre-redistricting boundary. The estimation sample requires at least one high-discretion permit during event times $t = -5$ through $t = -1$.

A.4 New Construction

The construction sample combines Cook County residential improvement records with commercial valuation records for properties assessed under multifamily model groups. I link records across parcel divisions, replacement PINs, and the two Assessor sources so that each project enters once. I then compare the resulting sample with every issued residential new-construction permit, adding completed projects absent from the Assessor files when units, building area, and lot area can be measured consistently.

Projects must have positive lot area, positive building area, at least one dwelling unit, and a construction year from 2006 through 2022. Assessor histories sometimes report different years or repeat whole-building characteristics across assessment cards. I use permit and completion dates to select the year that best matches completion, combine cards that clearly describe one building, and remove later PINs representing the same project. For 52 projects citywide, including 26 within 500 ft of a ward boundary, the administrative records require manual resolution; the replication files include a spreadsheet recording each decision.

Locations first come from the Historical Parcel Universe in the construction year. When the original PIN is unavailable there, I use the current location of that PIN or a verified historical address. Each project retains one location, preventing predecessor and successor PINs from entering separately. Multifamily status primarily follows Assessor property class. When the reported unit count does not plausibly describe the whole building, permit descriptions and other property records determine the whole-building count, with the remaining decisions recorded in the same spreadsheet.

Each project receives the broad zoning group in force on June 15 of its construction year. I combine official City zoning maps from 2012, 2014, and 2016 with passed map amendments dating to 2006, use annual parcel maps to follow sites whose PINs change, and retain planned developments as a separate zoning group. The reconstruction covers every project in the main and placebo boundary samples and reproduces the classifications observed in later official zoning maps. FAR is total building area divided by lot area; DUPAC

is dwelling units divided by lot acreage. Projects are assigned to the ward map and alderman in office on June 15 of the reported construction year because the Assessor does not provide an exact completion date. All ward assignment and distance calculations use EPSG:3435, with distance reported in feet.

A.5 New-Construction Summary Statistics

Table 1 describes the 3,692 projects within 500 ft used in both main density regressions, including 822 multifamily projects.

A.6 Home Sales

The sales sample covers Chicago transactions from 1999–2022, with the analysis restricted to 2006–2022. It includes residential classes 202–211 but excludes mixed-use properties and condominiums. Sales must exceed \$10,000, use a warranty or trustee deed, include recorded and non-identical seller and buyer names, and involve a single parcel; land-only transactions are excluded. Prices are deflated to 2022 dollars using the monthly Chicago CPI-U and winsorized at the 1st and 99th percentiles of the analysis sample.

I geocode each sale by matching its full PIN to the Historical Parcel Universe in the sale year, falling back to the 2025 location of the same PIN when necessary. Ward and alderman assignments use the exact sale date: transactions before May 18, 2015, use the old map, and those on or after that date use the new map. Signed distances are calculated in EPSG:3435 and oriented by relative aldermanic stringency.

A.7 RentHub Listings

The rental data begin as scrape-level listings. I standardize addresses and building types and treat zero, blank, NA, N/A, NULL, NONE, and UNKNOWN addresses as missing. Because the data do not identify units, a property proxy combines normalized address and

rounded coordinates, using coordinates alone when address is missing. A floorplan proxy adds bedrooms, bathrooms, square feet, and building type. I collapse repeated scrapes to the median rent for each floorplan and scrape date and then to the median daily rent for each floorplan-month. Before monthly aggregation, rent and square footage are trimmed at the 1st and 99th percentiles within year-by-bedroom cells.

The sample requires positive rent, valid coordinates, and assignment to a Chicago ward. Ward and alderman assignments use the listing's first observed date, with May 18 as the map transition date. The main analysis retains only reliable locations: it excludes manually rejected coordinates, unstable address clusters, addressless listings concentrated at generic coordinates, and cases where the address's modal coordinate changes the ward, neighboring ward, ward pair, or measured boundary distance by more than 100 ft. Regressions also require square footage, bathrooms, and amenity controls. Bedrooms enter categorically, so studios remain in the sample.

B Alderman Fixed Effects and Index Validation

This appendix provides additional detail on how the aldermanic stringency index is constructed and validated.

B.1 Construction of the Stringency Index

Aldermanic stringency is measured using processing times from the high-discretion permit sample described in Section A.2. The processing-time regression uses positive-duration permits through December 2022. Same-day permits remain in the lagged ward-month permit count.

Stage 1: Adjusting Processing Times. I regress permit-level log processing time on characteristics of the permit’s location (distance to the CBD, distance to the lake, and nearby CTA stations), ward demographic controls (median household income, homeownership rate, population, and racial composition), lagged ward-month permit volume, and fixed effects for year-month, permit type, and review type. Appendix A describes how block group demographics are allocated across ward boundaries to construct the ward controls.

Stage 2: Alderman Differences. I average the adjusted processing times by ward and month, then regress those ward-month averages on alderman indicators $\alpha_{a(w,m)}$ and the same lagged permit-volume control. Ward-months are weighted by their permit counts. Here $a(w,m)$ denotes the alderman serving ward w in month m . The alderman fixed effect captures the systematic component of processing time attributable to the alderman.

Empirical Bayes Shrinkage. Raw alderman fixed effects are shrunk toward the grand mean using an empirical Bayes procedure: $\hat{\alpha}_a^{\text{EB}} = B_a \hat{\alpha}_a$, where $B_a = \hat{\omega}^2 / (\hat{\omega}^2 + \hat{\sigma}_a^2)$ is the shrinkage factor and $\hat{\omega}^2$ is the estimated between-alderman variance.

Index Standardization. The final stringency index, S_a , is standardized to have mean zero and unit standard deviation across aldermen. Higher values indicate more stringent aldermen.

Table B.1 reports the first regression used to construct the stringency index. The dependent variable is permit-level log processing time. The specification controls for observed differences across wards and permit locations, as well as the mix of permit and review types. The purpose of this stage is to remove systematic differences in permit timing attributable to those observed characteristics before estimating the alderman-specific component. Panel B of Figure B.1 shows the distribution of the final stringency scores.

B.2 Validation of the Stringency Index

The stringency index is intended to capture alderman-specific regulatory frictions rather than purely ward-level conditions. Panel C of Figure B.1 compares predecessor and successor aldermen within the same ward. The weak relationship indicates that the index is not simply reflecting time-invariant ward characteristics.

Table B.1: First-Stage Processing-Time Regression

	Log Processing Time (1)
Median HH Income (\$10k)	-0.015 (0.016)
Share Black	0.175 (0.278)
Share Hispanic	0.401 (0.272)
Share White	0.529 (0.352)
Homeownership Rate	0.262 (0.222)
Population (10k)	-0.003 (0.020)
Dist. to CBD (km)	-0.017*** (0.006)
Dist. to Lake (km)	-0.002 (0.005)
CTA Stations (800m)	-0.009*** (0.003)
Lag Permits	-0.002*** (0.0006)
Year $\times$ Month FE	✓
Permit Type FE	✓
Review Type FE	✓
N	120,520

Notes: Permit-level OLS regression used to construct the stringency index, using permit records through December 2022. Covariates are median household income, racial shares, homeownership, population, distance to the CBD, distance to the lake, CTA station density, and lag permit volume. Fixed effects are listed in the table. Regressions are unweighted. Standard errors are clustered by ward and reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

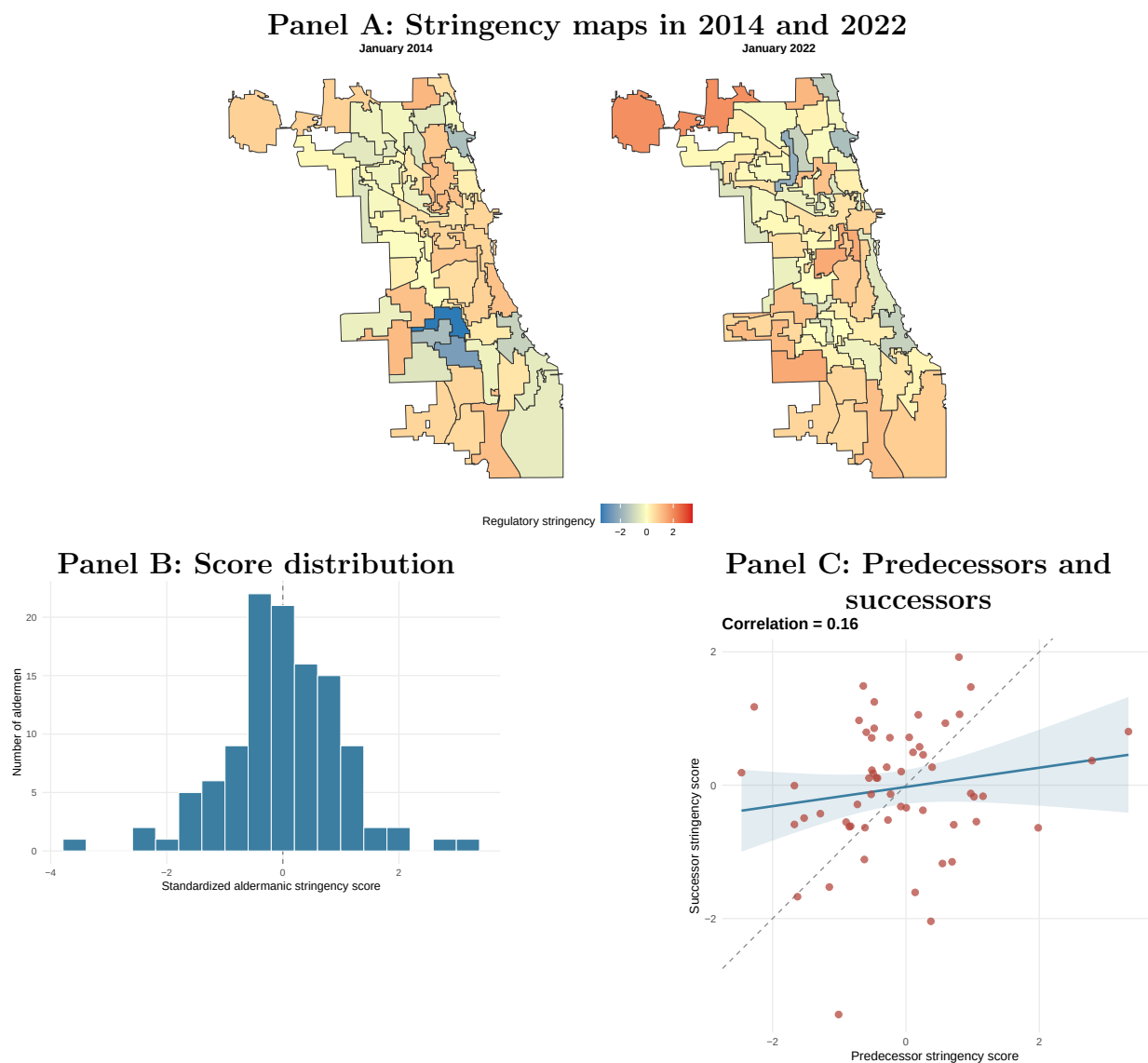


Figure B.1: Geography, Distribution, and Within-Ward Persistence of Aldermanic Stringency

Notes: Stringency scores are constructed from residualized permit processing times through December 2022 with empirical Bayes shrinkage and standardized to mean zero and unit standard deviation. Panel A maps the scores of aldermen serving in January 2014 and January 2022 onto the corresponding ward boundaries; both maps use the same color scale, with warmer shades indicating more stringent aldermen. Panel C compares each alderman's score with that of a predecessor or successor who represented the same ward in an adjacent period.

C Additional Permit Evidence

This appendix reports additional permit evidence that complements the descriptive discussion in Section 2 and the event study in Section 4.1.

C.1 Balance Before Redistricting

Table C.1 compares blocks that changed wards in 2015 with nearby blocks that remained in the same ward. Before the new map took effect, the two groups were broadly similar in permit activity and neighborhood characteristics. The joint test across all listed characteristics has a p -value of 0.333.

Table C.1: Pre-Redistricting Balance Between Reassigned and Unchanged Blocks

Variable	Control	Treated	Difference	SE	p -value	N
<i>Permit history</i>						
High-discretion permits, 2010–2014	2.668	2.649	0.086	0.081	0.291	2,340
Low-discretion permits, 2010–2014	4.278	4.377	0.146	0.412	0.725	2,340
No low-discretion permit, 2010–2014	0.114	0.127	0.008	0.023	0.713	2,340
New-construction permits, 2010–2014	0.381	0.349	0.013	0.044	0.766	2,340
No new-construction permit, 2010–2014	0.773	0.798	-0.008	0.013	0.525	2,340
<i>Neighborhood characteristics</i>						
Median household income	58,210	53,228	-1,352	1,354	0.324	2,324
Homeownership rate	0.404	0.407	0.000	0.011	0.998	2,340
Bachelor’s degree or higher share	0.441	0.399	-0.000	0.009	0.970	2,340
Black population share	0.301	0.272	0.003	0.006	0.667	2,340
Hispanic population share	0.226	0.276	-0.011	0.009	0.230	2,340
Median gross rent	1,068	1,022	-19	16	0.246	2,285
Median home value	309,400	283,787	-8,561	11,019	0.441	2,288
Average household size	2.550	2.694	-0.004	0.052	0.932	2,340
Median age	33.8	33.5	0.1	0.4	0.798	2,340
Population per square kilometer	8,350	8,187	-100	378	0.791	2,340

Notes: Treated blocks were reassigned to a different ward by the 2015 ward map; control blocks remained in their original ward. The sample requires the 2014 origin- and destination-ward incumbents to remain in office when the new map took effect. It also requires at least one high-discretion permit during 2010–2014. The sample contains 687 treated and 1,653 control blocks within 500ft of a ward boundary in 44 ward pairs containing both groups and is constructed before estimation. The event study can also compare blocks reassigned in opposite directions and therefore includes additional blocks. PPML drops blocks and ward-pair-year cells with no permits for the outcome being studied. Permits are grouped by application year. Neighborhood characteristics are 2014 ACS five-year block-group estimates. Difference is the coefficient on the treated indicator from a regression with ward-pair fixed effects; standard errors are clustered by ward pair. Means are unadjusted block-level means. The joint-test p -value for all listed covariates is 0.333 (complete-case $N = 2,249$).

C.2 Permit Event-Study Estimates

Table C.2 reports pooled estimates for years 0 through 5 after redistricting. It shows the main model, which treats moves toward greater stringency and greater leniency as opposite changes, and a more flexible model that estimates each direction separately. Low-discretion permits show a more muted response.

Table C.2: Permit Effects with Combined and Separate Reassignment Directions

	High-Discretion	Low-Discretion
Main estimate	-0.127** (0.049)	-0.017 (0.050)
<i>Directions estimated separately</i>		
Assigned toward more stringent	-0.231** (0.096)	0.037 (0.074)
Assigned toward more lenient	0.067 (0.059)	0.043 (0.065)
One-half difference between directions	-0.149*** (0.052)	-0.003 (0.048)
Equal-and-opposite test p -value	0.180	0.432
Block fixed effects	Yes	Yes
Ward-pair $\times$ year fixed effects	Yes	Yes
Positive pre-period permit activity required	Yes	Yes
N	36,830	35,541

Notes: Poisson pseudo-maximum likelihood estimates use census-block-year observations from 2010–2020 within 500 ft of a ward boundary. Stringency scores are estimated using permits filed from 2006 through 2014. The sample retains blocks whose 2014 origin- and destination-ward incumbents remained in office when the new map took effect. It also requires at least one high-discretion permit during event times $t = -5$ through $t = -1$. Blocks are classified as assigned toward greater stringency, assigned toward greater leniency, or unchanged under the new map. The main estimate constrains assignments toward greater stringency and greater leniency to have equal-and-opposite effects in logs. The second model estimates the two assignment effects separately. For the model that estimates each direction separately, the reported comparison is one-half of the difference between the more-stringent and more-lenient coefficients. The equal-and-opposite test evaluates whether the two separately estimated coefficients sum to zero. Each model includes block fixed effects and ward-pair $\times$ year fixed effects. Fixed-effect PPML drops blocks with no outcome in any sample year and ward-pair-year groups with no outcome. Standard errors, in parentheses, are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Allowing the Two Reassignment Directions to Differ

Panels B and C of Figure 2 allow moves toward greater stringency and greater leniency to have different effects. Table C.2 shows that one-half of the difference between those two effects is nearly identical to the main estimate.

Panel A of Figure C.1 shows the corresponding path for low-discretion permits. The response is more muted than for high-discretion permits.

C.4 Broader Assigned-Change Sample

Panel B includes blocks from wards that elected new aldermen in 2015. Treatment remains based on the 2014 origin- and destination-ward incumbents, so the estimate captures assigned rather than necessarily realized changes in stringency. The annual number of high-discretion permits is 8.3% lower, a somewhat smaller estimate than in the main sample.

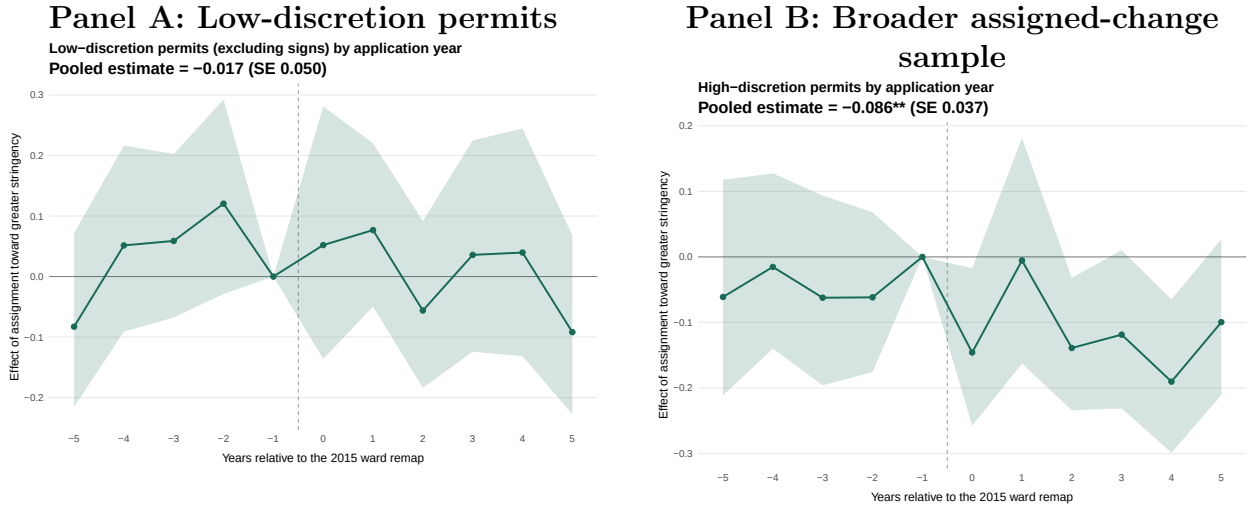


Figure C.1: Additional Permit Event-Study Checks

Notes: Panel A uses low-discretion permits excluding signs; its pooled log coefficient is -0.017 (SE = 0.050), with pre-trend $p = 0.066$. Panel B adds blocks from wards with aldermanic turnover while retaining treatment based on the 2014 incumbents; its pooled coefficient is -0.086 (SE = 0.037), with pre-trend $p = 0.665$. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

C.5 Permit Summary Statistics and Volume

Table C.3 compares the high- and low-discretion permit groups and reports their ward-month processing-time and permit-volume correlations. High-discretion permits are much slower than lower-discretion permits: the mean processing time is 57.3 days versus 26.2 days, and the median is 30 days versus 6 days. They also display more cross-alderman variation in processing times. Ward-months with longer high-discretion processing times have fewer high-discretion permits. Low-discretion permits show a weak positive relationship instead.

Panel A: Processing-time summary statistics

Statistic	High-Discretion	Low-Discretion
Total permits in sample	121,420	124,133
Mean processing time (days)	57.3	26.2
Median processing time (days)	30.0	6.0
Alderman-level mean of processing time	60.3	26.6
Alderman-level IQR of mean processing time	24.5	13.5
Number of unique wards	50	50
Number of unique aldermen	114	114

Panel B: Ward-month processing time and permit volume

Permit group	Corr. of mean processing time and ward-month permit volume
High-Discretion	-0.123
Low-Discretion	0.106
All	-0.021

Table C.3: Summary Statistics for High- and Low-Discretion Permits

Notes: The sample uses positive-duration permit records with applications begun through December 2022. The permit groups are defined in Section 2.2. The alderman-level interquartile range is computed across alderman-specific mean processing times. Panel B reports Pearson correlations across ward-month cells.

D Additional Density RD Checks

This appendix checks whether the density result appears at false cutoffs, reflects fixed differences in location, or depends on unusual boundary geometry or projects closest to the boundary. Unless otherwise noted, each estimate compares the two 100 ft bands immediately across the ward boundary using Equation (4).

D.1 Stringency-Score Checks

The first check asks whether a project's own permits influence the score used to classify its boundary. For each project, I recalculate the scores for the two aldermen after removing permits matched to that project. The estimates are effectively unchanged. The second check drops boundaries where the two alderman scores are close. All four point estimates remain negative when the minimum gap is either 0.25 or 0.50 standard deviations, although the multifamily estimates become less precise as the sample shrinks.

D.2 Placebo Cutoffs

Figure D.1 moves the cutoff 1000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority, and none of the eight nearest-band estimates is statistically distinguishable from zero.

D.3 Continuity in Location Characteristics

Table D.2 tests whether fixed features of a project's location change at the ward boundary. None of the differences in distance to downtown, the nearest school, the nearest park, or Lake Michigan is statistically distinguishable from zero in either sample.

Table D.1: Density Estimates Under Alternative Score Classifications

	All Construction		Multifamily	
	Log(FAR)	Log(DUPAC)	Log(FAR)	Log(DUPAC)
<i>Panel A: Removing permits linked to each project</i>				
Current score	-.129*** (.048)	-.172** (.067)	-.149** (.069)	-.194* (.099)
N	3,692	3,692	822	822
Ward pairs/segments	124/849	124/849	87/415	87/415
Score excluding the project's permits	-.129*** (.048)	-.172** (.067)	-.149** (.069)	-.194* (.099)
N	3,692	3,692	822	822
Ward pairs/segments	124/849	124/849	87/415	87/415
<i>Panel B: Excluding boundaries with similar scores</i>				
Minimum score gap: 0.25 SD	-.097* (.050)	-.140* (.076)	-.086 (.077)	-.097 (.103)
N	2,774	2,774	686	686
Ward pairs/segments	119/751	119/751	78/356	78/356
Minimum score gap: 0.50 SD	-.117** (.056)	-.160* (.086)	-.166 (.101)	-.188 (.130)
N	2,158	2,158	518	518
Ward pairs/segments	100/569	100/569	64/272	64/272

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The second row removes permits matched to each construction project before recalculating the alderman scores used for that project. The minimum-gap rows retain boundaries where the absolute difference between the two scores is at least the stated number of standard deviations. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

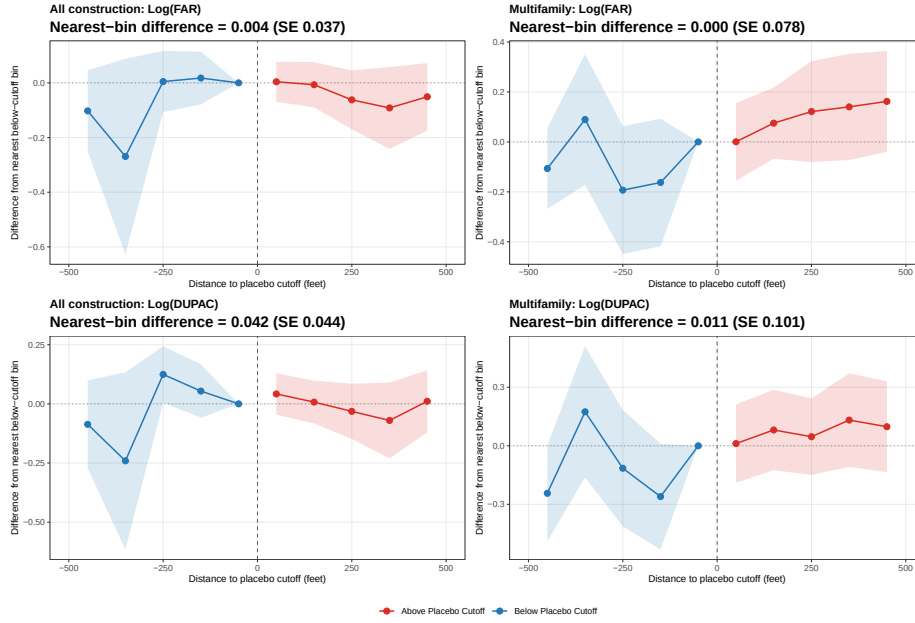
Table D.2: Continuity in Location Characteristics at Ward Boundaries

	All Construction	Multifamily
Distance to downtown (miles)	-0.001 (0.010)	-0.009 (0.012)
Distance to nearest school (miles)	0.005 (0.005)	-0.004 (0.009)
Distance to nearest park (miles)	-0.005 (0.006)	-0.010 (0.010)
Distance to Lake Michigan (miles)	0.004 (0.007)	-0.003 (0.011)
Observations	3,692	822

Notes: Each row uses the listed location characteristic as the dependent variable. The reported coefficient compares the 0–100 ft band on the more-stringent side with the –100 to 0 ft band on the less-stringent side. Regressions include 100 ft distance-band indicators, boundary-segment fixed effects, and construction-year fixed effects. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: 1000 ft inside the less-stringent side

Placebo cutoff 1,000 feet inside the less-stringent ward



Panel B: 1000 ft inside the more-stringent side

Placebo cutoff 1,000 feet inside the more-stringent ward

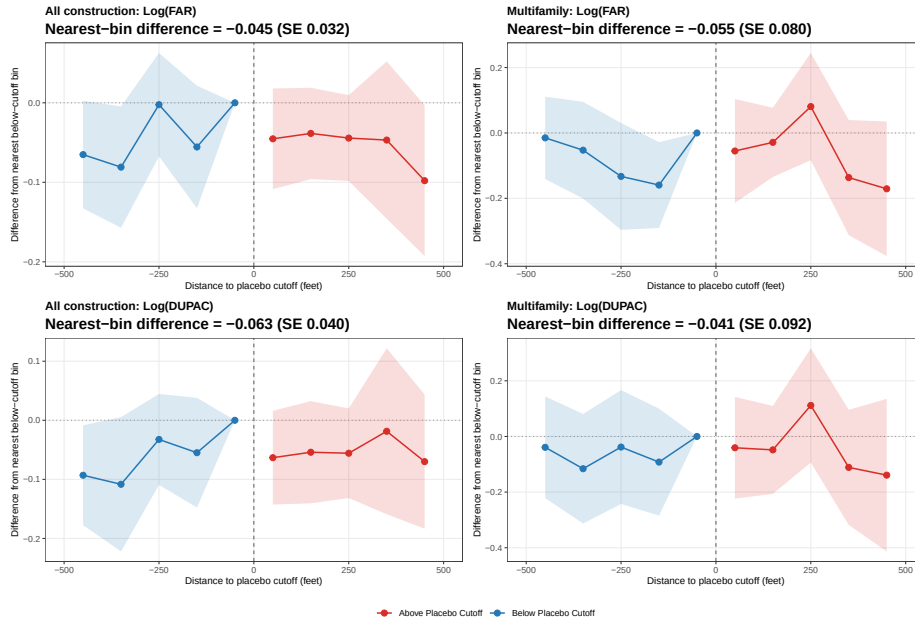


Figure D.1: Density Estimates at Placebo Cutoffs

Notes: Each artificial cutoff lies 1000 ft inside the indicated side of the true ward boundary. The model uses a 500 ft window around the artificial cutoff and the same 100 ft distance bands, controls, and fixed effects as Equation (4). The nearest band below the artificial cutoff is omitted and normalized to zero; subtitles report the difference in the nearest band above it. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

D.4 Boundary Geometry

Some ward boundaries follow expressways, waterways, arterial roads, or other physical features. Table D.3 therefore repeats the analysis after restricting the sample based on the geometry of the assigned boundary segment. Removing segments with substantial expressway or water overlap leaves the estimates almost unchanged. A broader physical-feature and arterial-road restriction attenuates the multifamily estimates, while the all-construction estimates remain negative. Restricting the sample to locally straight boundaries also produces four negative estimates. These estimates are generally smaller and less precise in the reduced sample, but they do not suggest that boundary curvature alone creates the main pattern.

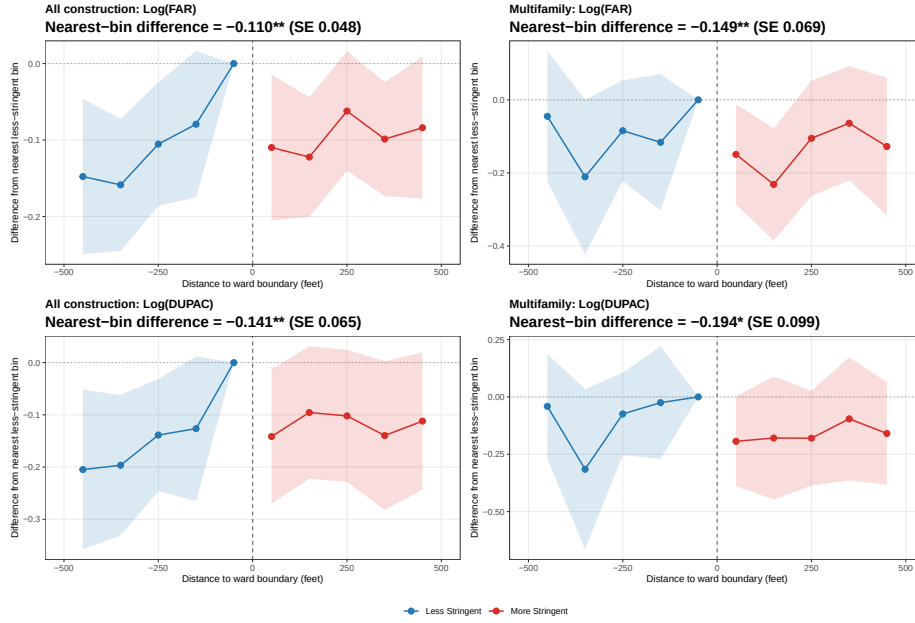
Table D.3: Density Estimates Under Boundary-Sample Restrictions

	All Construction		Multifamily	
	Log(FAR)	Log(DUPAC)	Log(FAR)	Log(DUPAC)
Main sample	-0.129*** (0.048)	-0.172** (0.067)	-0.149** (0.069)	-0.194* (0.099)
Observations	3,692	3,692	822	822
Limited expressway or water overlap	-0.128*** (0.047)	-0.171** (0.067)	-0.149** (0.069)	-0.194* (0.099)
Observations	3,443	3,443	799	799
Limited physical-feature or arterial overlap	-0.118** (0.046)	-0.160** (0.062)	-0.112 (0.070)	-0.106 (0.088)
Observations	2,888	2,888	660	660
Straight boundary segment	-0.085 (0.052)	-0.125** (0.059)	-0.096 (0.082)	-0.173* (0.101)
Observations	2,877	2,877	592	592

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The expressway-or-water restriction removes segments for which either feature overlaps at least half of the segment. The broader restriction removes segments with at least 50% park, water, or cemetery overlap, 40% expressway overlap, or 75% arterial-road overlap. A boundary is locally straight when both endpoints of a 328 ft line tangent to the project's nearest boundary point fall within 49 ft of the actual boundary. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Excluding projects within 25 ft

True ward boundary, excluding projects within 25 feet



Panel B: Excluding projects within 50 ft

True ward boundary, excluding projects within 50 feet

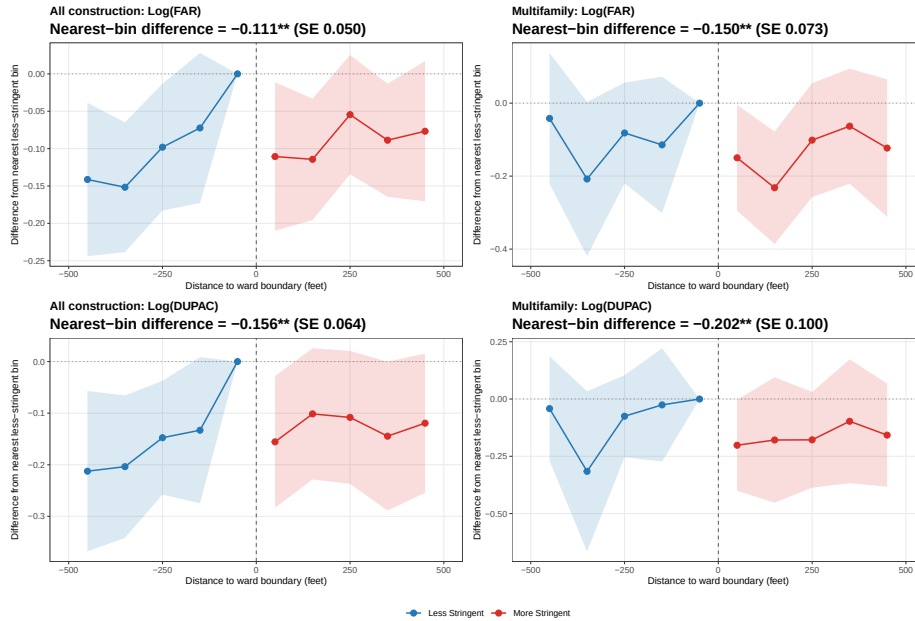


Figure D.2: Density Estimates Excluding Projects Nearest the Boundary

Notes: This check asks whether projects immediately adjacent to a boundary drive the main result. Each panel excludes projects within the indicated distance and otherwise uses the 500 ft sample and Equation (4). After each exclusion, the remaining portion of the -100 to 0 ft band forms the omitted category. In each panel, all four nearest-band estimates remain negative. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

E Additional Price Boundary Checks

Shifted-cutoff price placebos

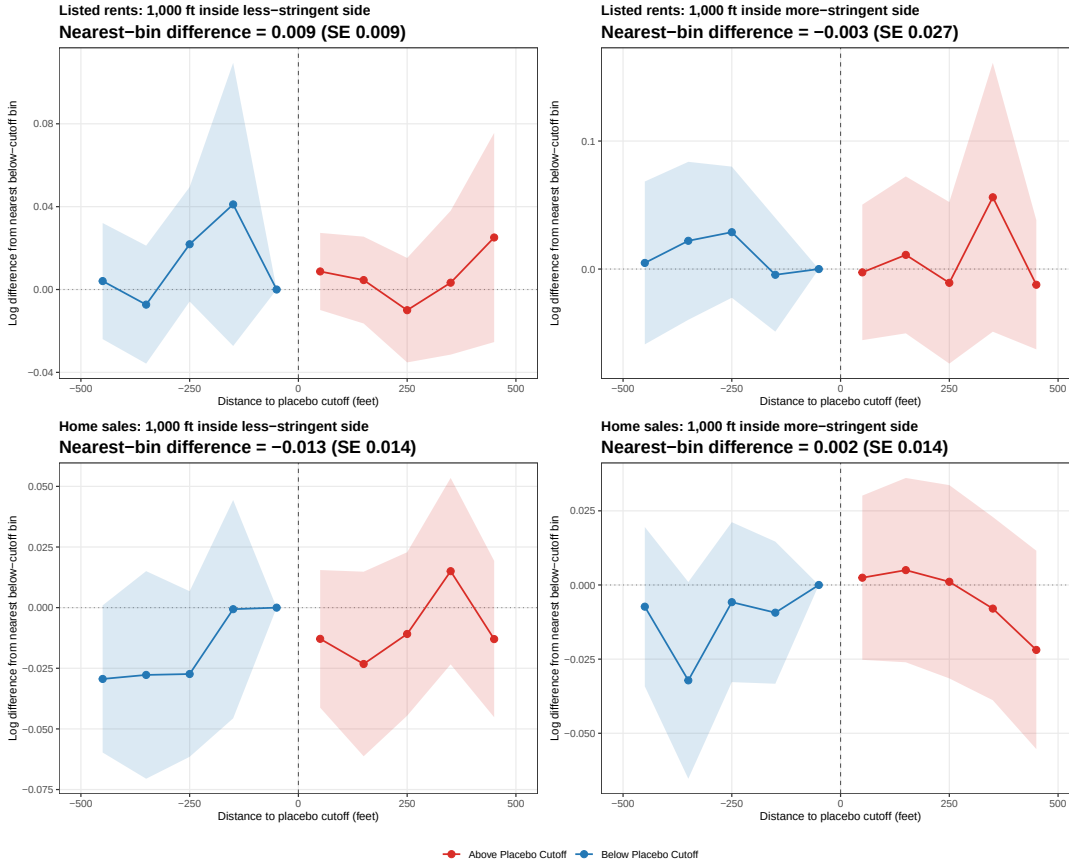
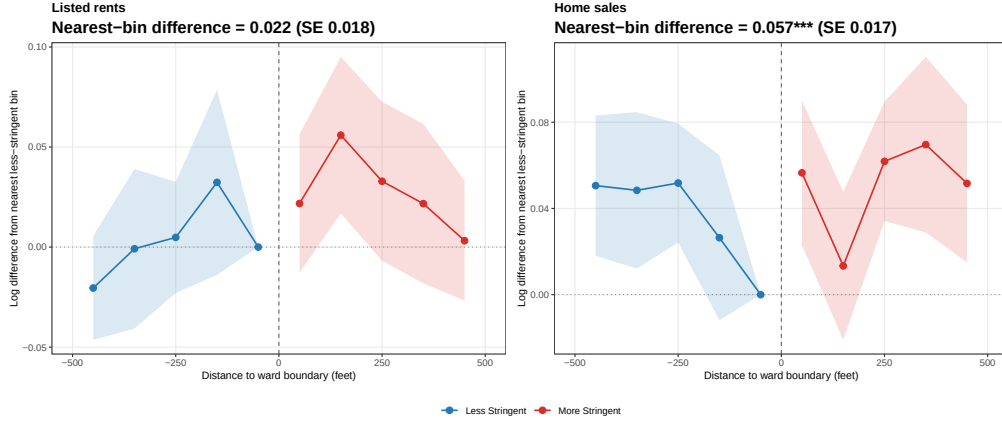


Figure E.1: Placebo Cutoffs for Listed Rents and Home Sale Prices

Notes: The placebo windows do not overlap the main 500 ft sample, and none of the nearest-band estimates is statistically distinguishable from zero. The left column places an artificial cutoff 1000 ft inside the less-stringent ward; the right column places it 1000 ft inside the more-stringent ward. The top row reports listed rents and the bottom row reports home sale prices. Each model uses a 500 ft window around the artificial cutoff and otherwise follows the corresponding specification in Figure 4. Each point is the estimated log difference between a 100 ft distance band and the band immediately below the artificial cutoff. Shaded areas are 95% confidence intervals. Standard errors are clustered by ward pair.

Panel A: Locally straight boundaries

Price estimates near locally straight ward boundaries



Panel B: Donut samples

Price estimates excluding observations nearest the boundary

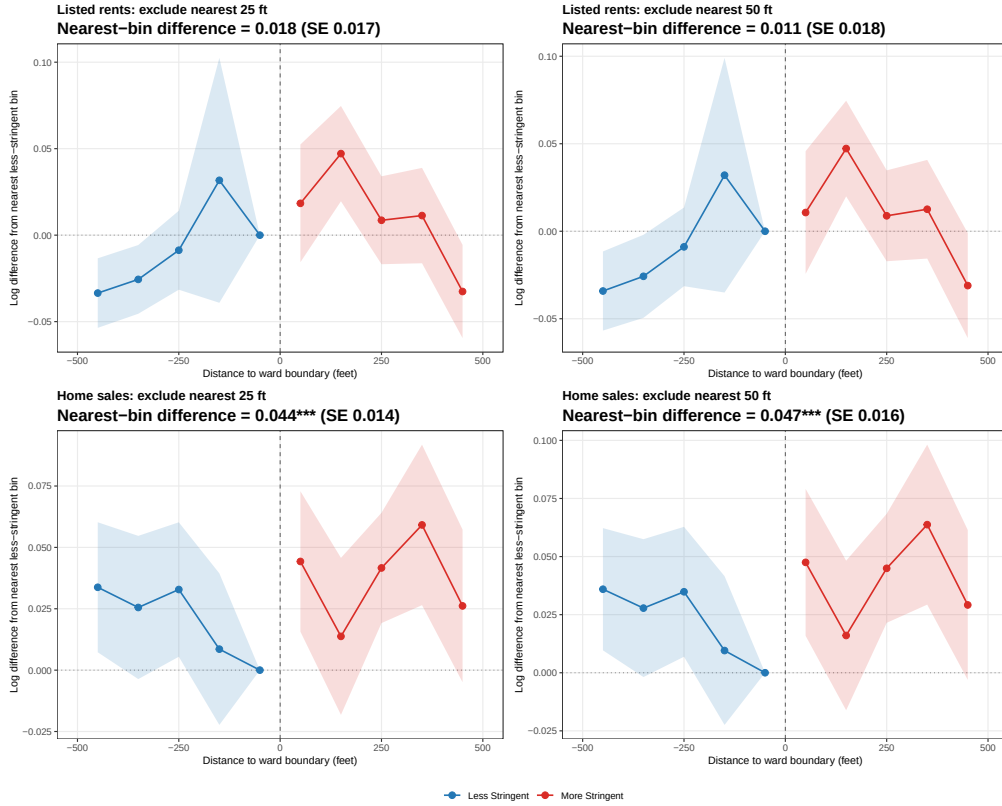


Figure E.2: Additional Price-Boundary Robustness Checks

Notes: Panel A restricts the sample to locally straight boundaries; its estimates are somewhat larger than the full-sample results. It retains locations where both endpoints of a 328 ft line tangent to the nearest boundary point lie within 49 ft of the actual boundary. In Panel B, the left column excludes observations within 25 ft and the right column excludes observations within 50 ft; the top row reports listed rents and the bottom row home sale prices. The donut estimates are similar to the full-sample results. All models otherwise follow Figure 4. Shaded areas are 95% confidence intervals, and standard errors are clustered by ward pair.